

A Smoothed-Particle Hydrodynamics Visualization of a Grazing Mars–Earth Collision

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ABSTRACT

We describe a numerical and rendering workflow for a near-grazing collision between differentiated Earth- and Mars-mass bodies. The calculation was developed to support a scientific visualization of Solar System instability outcomes, rather than to establish a converged impact outcome. Initial bodies are generated with WoMa, relaxed in SWIFT, assembled on a grazing trajectory, and evolved with self-gravity and hydrodynamics. Surface tracer particles are advected with the flow and colored to indicate terrestrial land/ocean material and the Mars impactor. The current high-resolution visualization uses 218,271 SPH particles, an impact angle of 70 degrees from head-on, and a contact velocity of 1.02 times the mutual escape speed. The animation shows strong tidal deformation of the impactor, a grazing exchange of material, and a departing Mars remnant embedded in a debris stream. We emphasize the distinction between this presentation-quality simulation product and a publication-quality convergence study: future work must quantify energy and angular-momentum conservation, equation-of-state sensitivity, resolution dependence, and outcome boundaries in velocity-angle-impact-parameter space.

Keywords: Hydrodynamical simulations (767) — Impact phenomena (779) — Terrestrial planets (1708)
— Planetary collisions (2203)

1. INTRODUCTION

The long-term stability of terrestrial planetary systems is punctuated, in some regions of parameter space, by close encounters and direct collisions. For public and scientific presentations, these events are difficult to convey with point-mass integrations alone: the dynamical encounter is driven by orbital mechanics, but the memorable observable consequence is a material process involving tidal distortion, ejecta, accretion, and possible survival of a coherent remnant. The present project builds a reproducible local workflow for visualizing one such event, a grazing encounter between Earth- and Mars-mass bodies.

The calculation is not intended as a claim that a specific Mars–Earth impact is likely in the present Solar System, nor as a converged prediction for the outcome of any particular instability scenario. Instead, it provides a scientifically grounded animation: the bodies are differentiated, self-gravity is active, material surfaces are advected as tracers, and the rendering is tied to the actual SPH particle state. This document records the numerical setup, the rendering strategy, and the limitations that separate the current product from a peer-reviewed impact-outcome survey.

2. NUMERICAL SETUP

The initial conditions are generated with WoMa and evolved with SWIFT. The target has approximately Earth mass and radius, while the impactor has approximately Mars mass and radius. Both bodies are differentiated iron-silicate objects using ANEOS Fe85Si15 and forsterite material tables. Separate Earth and Mars bodies are first relaxed, then assembled into an impact configuration.

The main rendered calculation uses a requested resolution of $N_{\text{total}} = 200,000$, yielding 218,271 particles after shell population adjustments and subsequent particle handling. The nominal impact geometry is a 70 degree angle from head-on, with an initial contact speed of $1.02 v_{\text{esc}}$, where v_{esc} is the two-body mutual escape speed. The trajectory is initialized about two hours before nominal contact and is continued to nine hours of simulated time. The exact configuration files, diagnostics, compact initial-condition products, and logs are included in the repository.

3. RENDERING METHOD

Visualization is performed from the SPH particle snapshots. Each particle is matched to a sidecar label file by its **ParticleIDs** value. Label files store the source body, surface class, longitude/latitude at initialization, and RGB colors. The Earth surface col-

ors represent a coarse present-day land/ocean pattern, with a rendering-only coastline refinement that demotes continental-shelf-adjacent particles from land to ocean for a cleaner visual map. The Mars colors use a procedural albedo pattern intended to evoke the global appearance of Mars, including ochre dust, darker basaltic provinces, and subtle polar frost.

The direct-view movie is rendered in an inertial frame. The camera is fixed, but the projected screen axes are chosen so the final Earth–Mars separation is nearly horizontal, allowing the last frame to use the 16:9 aspect ratio efficiently. Bounds are set by robust final-frame Earth and Mars extents, excluding distant stray particles. Surface tracers are drawn as crisp depth-sorted points over a dim body-base layer, while the storyboard and density-trial renderers provide alternative views for presentation work.

4. CURRENT OUTCOME

The animation displays a strongly grazing event rather than a direct head-on merger. During the close interaction, the Mars-mass impactor is tidally elongated and partially stripped. Some Mars material is transferred into a debris stream and onto/near the target, while a coherent Mars-rich remnant continues on a departing trajectory. The Earth surface tracer field is distorted and contaminated by impactor material near the encounter region. By nine hours, the two main mass concentrations are well separated in the inertial-frame render, with scattered inter-body debris visible between them.

The visual result is useful for communicating the physical distinction between a point-mass near miss and a hydrodynamical grazing collision. In particular, the impactor does not simply rebound as a rigid body: it is reshaped, stripped, and connected to the target by an extended material wake. At the same time, the calculation should not yet be interpreted as a quantitatively robust boundary between hit-and-run, merger, and erosion regimes.

5. LIMITATIONS AND REQUIRED PUBLICATION WORK

Several steps remain before this workflow can support publication-quality physical conclusions. First, the resolution ladder must be extended and analyzed quantitatively. The present 200k-particle render is suitable for visualization, but it is low by the standards of detailed giant-impact outcome studies. Second, conservation diagnostics must be reported systematically for the production snapshots: total energy, angular momentum, linear momentum, mass partitioning, and thermodynamic state should be tracked as functions of time.

Third, the initial thermal profile, differentiated structure, equation of state, and relaxation procedure should be varied to assess sensitivity. Fourth, any outcome classification should be embedded in a grid over impact angle, velocity, mass ratio, spin state, and internal structure.

The rendering pipeline also imposes interpretive choices. Surface tracers are faithfully advected particle labels, but the land/ocean and Mars albedo maps are visual encodings, not resolved geology. The density-layer trial renderer uses the SPH smoothing lengths to create a projected semi-transparent field, which is useful for seeing interiors and debris curtains but is still a visualization product rather than a radiative-transfer calculation.

6. REPRODUCIBILITY

The GitHub repository stores the scripts, compact initial-condition files, sidecar labels, YAML configuration files, diagnostics, logs, rendered movies, and quality-control stills. The largest generated snapshot directories are not stored directly in Git; they are documented in `manifests/LARGE_ARTIFACTS.md`. Recreating the full rendered sequence requires the SWIFT and WoMa environments documented in the repository, the corresponding snapshot time series, and the renderer commands recorded in the README and scripts.

7. SUMMARY

We have assembled a local, reproducible workflow for a visually compelling Mars–Earth grazing-collision animation. The current result combines differentiated SPH bodies, a high-resolution settled impact run relative to the exploratory ladder, advected surface tracers, inertial-frame camera choices, and presentation-oriented render products. The output is appropriate as a scientifically informed animation asset and as a foundation for a future quantitative study. Publication-quality impact claims will require a broader resolution and parameter survey with explicit conservation and convergence tests.

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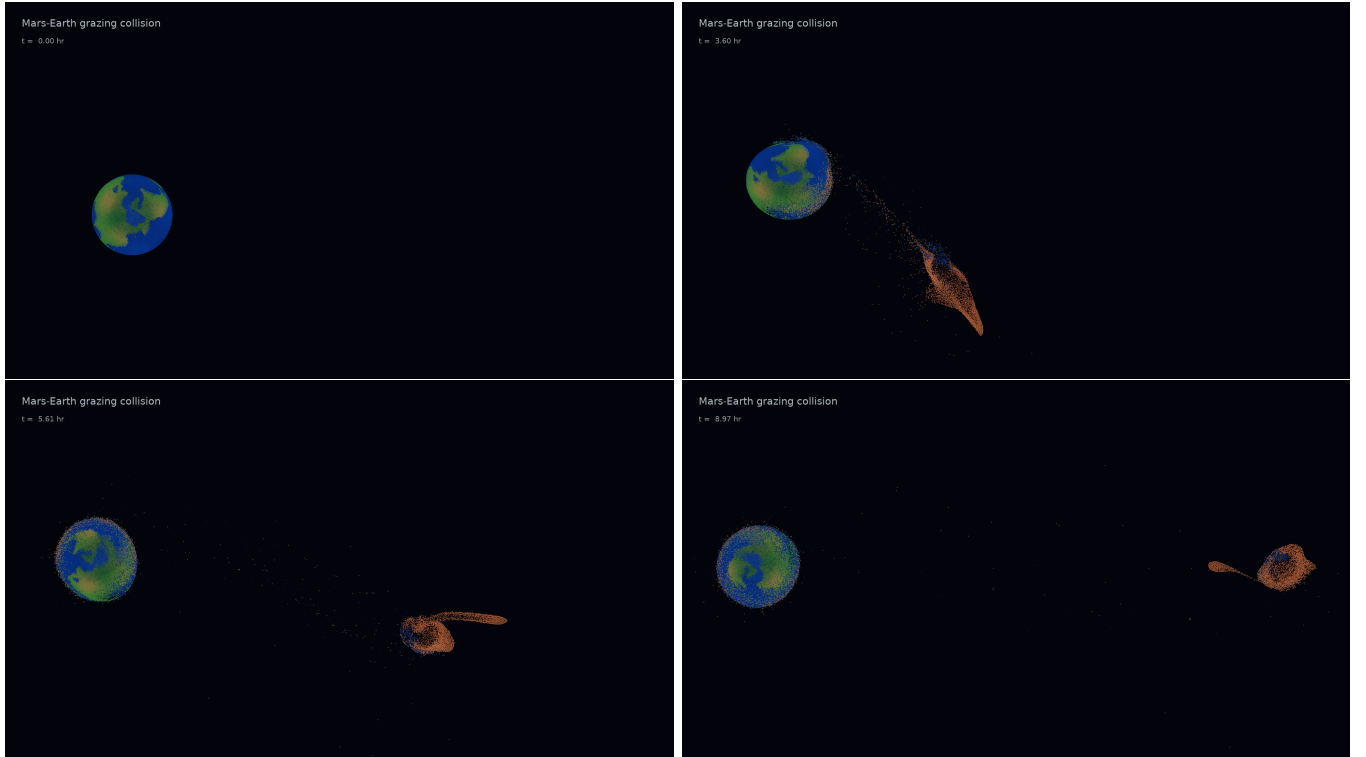


Figure 1. Selected frames from the inertial-frame direct-view render. The panels show the initial configuration, close interaction, departure, and final nine-hour state. The surface colors are tracer labels advected with SPH particles, not texture maps pasted onto a rendered sphere.